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What is an Integer Linear Program?

NP-Hardness of Integer Linear Programs

Tutorial: Solving Integer Linear Programs using Python/Pulp

Vertex Cover as Integer Linear Program

Linear Programming Approximations of Vertex Cover

Solving Integer Linear Programs using Branch and Bound Algorithm

Video: Branch and Bound Algorithm for Solving Integer Linear Programs
27 min

Lab: Interactive Notes: Branch and Bound Solvers for ILPs
1h

Quiz: Branch and Bound Solvers
30 min

Problem Set # 2

Branch and Bound Solvers

Review Learning Objectives

Submit your assignment

Due Feb 18, 11:59 PM IST

Receive grade

To Pass 80% or higher

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1. Suppose we are solving an ILP using branch and bound algorithm. Select all the facts below about the algorithm that are true. 6 / 6 points

- ☐ If a particular LP that we solve during branch and bound is unbounded, the original ILP can still be bounded
- ☒ If the LP relaxation at the "root" of the branch and bound is infeasible, the whole ILP must be infeasible.

☒ Correct

This is correct because at the root, we are solving the LP relaxation of the original problem.

☐ If a particular LP that we solve during branch and bound is infeasible, we conclude that the original ILP is infeasible.

☒ Each step of branch and bound solves the LP relaxation of some ILP problem that is either the original problem or obtained by branching.

☒ Correct

Correct.

☐ Suppose the original ILP seeks to minimize an objective. If we found an integer solution during branch and bound, its objective value is guaranteed to be less than or equal to the actual ILP solution.

☒ If the original ILP is unbounded, then the LP relaxation at the root of branch and bound (the very first LP we will solve) will be unbounded.

☒ Correct

Correct since it is a LP relaxation of the ILP itself.

2. We are solving a branch and bound problem and at some stage, we solve a LP relaxation that yields the solution $x_1 = 2.5, x_2 = 1.3, x_3 = 5, x_4 = 6.9$. 4 / 4 points

Select all the valid options for the branch and bound algorithm.

- ☐ We recursively solve two subproblems with the constraints $x_3 \leq 4$ and $x_3 \geq 6$, respectively.
- ☐ We may recursively solve two subproblems with the constraints $x_2 \leq 2$ and $x_2 \geq 1$, respectively.
- ☐ We can stop and return the solution as an optimal solution for the ILP.
- ☒ We recursively solve two subproblems with the constraints $x_4 \leq 6$ and $x_4 \geq 7$, respectively.

☒ Correct

Correct: this is a valid move as far as the b&b algorithm goes.

☒ We may recursively solve two subproblems with the constraints $x_1 \leq 2$ and $x_1 \geq 3$, respectively.

☒ Correct

Correct: this is a valid move as far as the b&b algorithm goes.

3. We are solving an ILP whose objective function is $\max x_1 + x_2 - x_3$. At some point in the branch and bound algorithm, we have seen integer solutions shown below: 4 / 5 points

x_1	x_2	x_3	objective
1	2	-1	4
4	1	2	3
5	2	-2	9
1	1	1	2

Select all true facts from the list below.

- ☐ The optimal solution of the ILP will have objective value less than or equal to 2.
- ☐ Suppose we solve an LP relaxation and find that the objective value is 9.8 , we can prune this branch from further consideration.
- ☒ Suppose we solve an LP relaxation and find that the objective value is 8.5, we can prune this branch from further consideration.

☒ Correct

Correct: any integer solutions we obtain by expanding this branch further must have objective ≤ 8.5 , but we already have an integer solution with objective of 9

☒ The optimal solution of the ILP will have objective value greater than or equal to 9.

☒ Correct

Indeed: this is correct since we are solving a maximization problem.

☐ Suppose we solve an LP relaxation and find that the objective value is 10.2, we can prune this branch from further consideration.

You didn't select all the correct answers

Try again

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